

OM DEHLAN

+1 (919) 923 0719 ♦ om.dehlan@gmail.com ♦ [GitHub](#) ♦ [LinkedIn](#)

EDUCATION

University of North Carolina at Chapel Hill

Expected May 2027

Master of Science in Computer Science

CGPA: 4/4

Courses: Game Theory in Learning, Multimodality in LLMs, Advanced Topics in NLP, Transfer Learning

Indian Institute of Technology Delhi (All India Rank 73, JEE Advanced 2021)

May 2025

B.Tech. in Computer Science and Engineering, and Specialization in Data Analytics and AI

CGPA: 8.51/10

Courses: Deep Learning, Natural Language Processing Calculus, AI, ML, Abstract Algebra, Linear Algebra and Differential Equations, Probability and Stochastic Processes, Approximation Algorithms

ACADEMIC AND SCHOLARSHIP ACHIEVEMENT

- **JEE Advanced:** Achieved **All India Rank 73**, in engineering entrance process of **1,000,000+** applicants.
- **Indian National Mathematics Olympiad:** Qualified for INMO 2021 as **top 30** scorer in all of 12th grade.
- Indian Olympiad Qualifier in Astronomy 2020-21: Awarded the certificate of Merit as **State Topper** for having been placed among **Top 1%** candidates of Delhi.
- **Scholarships & Fellowships:** Recipient of the NTSE Scholarship, KVPY Fellowship (All India Rank **517** in KVPY-SX 2021 among **100,000+** students), IIT Delhi Endowment Merit Fellowship (2021-22), and IIT Delhi Semester Merit Award for being among the **Top 7%** of the batch.

TECHNICAL SKILLS

Languages:

Python, C, C++, SML/NJ, Prolog, MIPS (Assembly), VHDL, Bash, MySQL, HTML5, CSS, JS

Software, Tools and Libraries:

PyTorch, PEFT, CUDA, LangChain, AutoGen, OpenMP, MPI, vLLM, HuggingFace Transformers, Git, Vivado

PUBLICATIONS

[LRPlan](#): Accepted at Findings of EMNLP 2025. Suzhou, China. November 2025 [Aug'24 - May'25] (Prof. Mausam and Dr. Manish Gupta)

- Co-first-authored LRPLAN, a domain-independent multi-agent framework where LLM and LRM agents collaborate for constrained planning, achieving **SoTA** on TravelPlanner and TimeArena-Static with up to **12.7 pt** accuracy gains over nearest baselines while maintaining superior cost efficiency with code made public [here](#).

EXPERIENCE

Research Intern (Prof. Hyung Ju Hwang)

May 2024 - July 2024

Pohang University of Science and Technology

Pohang, Republic of Korea

- Built arbitrary-scale **super-resolution** models using **diffusion**, **transformers**, and **neural operators**
- Developed real-time **3D object detection** pipeline fusing LiDAR and RGB on KITTI and nuScenes.
- Deployed a **RAG chatbot** with Mixtral-8x7B, **LangChain**, and SQL-backed retrieval in **PyTorch**.

PROJECTS

- **Semantic Mode Collapse in LLMs** Prof. Snigdha Chaturvedi **Present**
 - Developing training-time and decoding-level methods to improve **semantic diversity** in LLM outputs using **reinforcement learning** and explainable diversity rewards.
- **Slot Learning with Object-Centric Diffusion** Prof. Parag Singla **Apr'24 – May'24**
 - Implemented **slot attention** and **diffusion decoder** in **PyTorch** for object-centric **representation learning** and compositional **image generation**.
- **Rollerball Game Bot** Prof. Mausam **Oct'23 – Nov'23**
 - Built a **C++** game-playing agent using **Minimax** with **alpha-beta pruning** and **Monte Carlo Tree Search (MCTS)** with learned evaluation functions.

EXTRA-CURRICULAR RESPONSIBILITIES AND VOLUNTEERING

- **Diversity Representative** (Office of Diversity and Inclusion, IIT Delhi); **Administrative Coordinator** (Vibhinn 2024; IIT Delhi Pride Fest); **Academic Mentor, Calculus and Student Mentor** (Board for Student Welfare, IIT Delhi); **Logistics Director and Hostel Representative, Dance** (Board for Recreational and Creative Activities, IIT Delhi); **Hospitality Executive, Inter-IIT Sports Meet** (Board for Sports Activities).